



Entrepreneurs as designers of problems worth solving

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ABSTRACT

Finding meaningful problems to address is a critical driver of innovation and entrepreneurship in today's turbulent environment, possibly even more so than problem solving. While the literature has mostly investigated problem finding in terms of discovering hidden user needs or root causes, this paper takes a different approach, namely problem finding as a design act. In particular, the paper introduces key concepts and methods for designing problems worth solving predicated on the practice of reframing, thus creating novel frames to look at a situation from an original perspective and make new sense of it.

1. Introduction

There is growing interest in how innovators and entrepreneurs find meaningful problems to address in order to create novel value (Nickerson et al., 2012; von Hippel & von Krogh, 2016). While traditionally the focus has been on problem solving (Jeppesen & Lakhani, 2010; Sommer et al., 2020), problem finding – here defined as identifying and formulating problems (Baer et al., 2013; Jay, 1996) – is receiving extensive attention for multiple reasons. First, the pace of technological progress and societal change accelerates the obsolescence of existing problems and the generation of new ones (Lai & Grønhaug, 1994), for example, fatigue from intense remote working during the pandemic. Second, while problem solving is increasingly delegated to artificial intelligence, problem finding remains inherently human. Indeed, personal purpose and values largely influence whether a problem is deemed worthy of effort (Csikszentmihalyi, 1988; Verganti et al., 2020). Third, the wicked nature of problems characterized by paradoxical elements, blurry boundaries, and conflicting stakeholders (Rittel & Webber, 1973), calls for advanced problem formulation skills, a prerequisite to generate “both... and...” solutions that combine innovation and sustainability, profit and purpose (Mitchell & Walinga, 2017).

Management scholars have articulated concepts and practices to find problems worth solving. These include empathic, ethnographic research techniques to unearth hidden user needs (Kelley & Kelley, 2013; Liedtka, 2015), analytical methods to decompose the problem, deep-diving into its root causes (von Hippel & von Krogh, 2016), and zeroing-in on the job to be done (Christensen et al., 2016). These cues help reach the moment of clarity (Madsbjerg & Rasmussen, 2014) wherein the entrepreneur identifies the fundamental essence of the problem that drives its subsequent formulation and solution.

Although relevant, these studies adopt a narrow view of problem finding as the discovery of problems “out there” by analyzing their key antecedents. This retrospective orientation overlooks emerging so-

cial, cultural, technological shifts that can render the problem and its structure obsolete. It confines the role of the entrepreneur to that of a detective inquiring into and reacting to the needs of current users (Verganti, 2017), a limiting role that hinders the potential of problem finding as a key driver of novel value creation.

Therefore, this paper adopts an alternative perspective to problem finding, namely designing more meaningful problems. Indeed, problems can also be created or invented, not only solutions (Getzels, 1979; Lai & Grønhaug, 1994).

This perspective draws on several key notions. First, a broad definition of problems as gaps between what is today and what might be in the future (Boland et al., 2008). At any point in time, a domain of any sort, be it an area of human life or a product category, presents gaps, regardless of whether the current situation is perceived as acceptable or harmful and unsatisfactory¹ due to unmet needs and pain points (Kepner & Tregoe, 1975; Landry, 1995). According to Dewey (1920), “...the specific conditions which exist at one moment, be they comparatively bad or comparatively good, in any event may be bettered” (pp. 181–182). Even when the current situation is deemed satisfactory without apparent tensions, the realization of a desired future state presents barriers that are essentially problems to be solved (Felin et al., 2020). In this broad connotation, the notion comprises opportunity vision, a foundational artifact in entrepreneurship research (Berglund et al., 2020).

Second, a design perspective to problem finding entails a more generative role of entrepreneurs as visionaries who forge novel problems by exploring what is possible tomorrow (Hekkert & van Dijk, 2011; Van der Bijl-Brouwer & Dorst, 2017). In the hypothetical new world shaped by ongoing and future environmental forces, what experiences will be meaningful to people and how can we offer them? Behind this open-ended, prospective sensemaking (Stigliani & Ravasi, 2012) is the

¹ This corresponds with the classic view of problems as felt difficulties (Dewey, 1910), a view focused on negativities (Verganti, 2017).

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typical approach of professional designers expanding the functionality of any domain “to include an enlarging circle of concern for emotions, customer experiences, ethical behavior, environment, cultural norms, and esthetic appeal” (Boland et al., 2008, p. 23).²

Third, frames are a central concept of designing problems worth solving. Unlike machines, humans subjectively frame the reality they perceive (Cukier et al., 2021; Goffman, 1974), interpreting it by creating a mental representation (Cornelissen & Werner, 2014). Like any model, frames selectively direct attention (Dimov, 2016; Leonardi, 2011), acting as lenses that determine which aspects of the situation are in focus and which are not. Designing novel problems is essentially an act of reframing, creating new lenses to make sense of the situation in an original way, shifting the focus to alternative aspects that were not part of the initial problem scope but are then creatively brought in by the entrepreneur (Dorst, 2015). The newly designed frame suggests a direction to expand the domain’s functionality, specifying the desired nature of “what it might be”.

The much-cited Apple iPhone is an exemplary case of problem finding as design. This product was not developed as a solution to an existing problem of mobile phones, but to propose radically new experiences. Through building novel mobile communication frames that differed from the conventional, Apple injected new needs into the domain that users had not yet perceived, to the extent that its functionality evolved from being mainly about connecting people to running their lives by way of a pocket-sized device. The genesis of other innovations, such as Nintendo Wii and Cirque du Soleil (Kim & Mauborgne, 2019), epitomize the problem finding design perspective.

Here we articulate the main concepts and practices related to designing novel and more meaningful problems by integrating insights from design (Dorst, 2015; Hekkert & van Dijk, 2011), entrepreneurship (Felin et al., 2020; Zellweger & Zenger, 2021), innovation (Christensen et al., 2016; Verganti, 2017), and strategy (Kim & Mauborgne, 2014; Ramirez & Wilkinson, 2016). In so doing, we make two key contributions. The first is to the management (Beckman, 2020; Liedtka, 2015) and entrepreneurship literature (Berglund et al., 2020; Dimov, 2016) that takes a design perspective of organizational phenomena. Seeing problem finding as an act of design and not just discovery expands the potential impact of design practice from the creative generation of solutions to that of problems. Speculatively reframing problems is an additional lever for breakthrough innovation, since an unconventional problem formulation can open up unexpected solution paths. This supports a more strategic role of design in business (Van der Bijl-Brouwer & Dorst, 2017).

The second contribution is to general entrepreneurship research and practice, which currently tends to focus more on how to execute visions rather than imagining them in the first place. While lean startup studies investigate the use of minimum viable products and smoke tests to validate vision-derived hypotheses (Eisenmann et al., 2013; Leatherbee & Katila, 2020), we propose an approach to developing novel visions through problem reframing. This helps counter the risk that an excessive focus on execution cuts entrepreneurship off from the central strategic stage linked to envisioning meaningful directions in the future, amongst others.

2. Problem finding in the literature

The importance of problem finding for innovation has long been emphasized, also by great thinkers of the past, “To raise new questions, new problems, to regard old problems from a new angle, requires creative imagination and marks real advances in science” (Einstein & Infeld, 1938, p. 92). Some even argued for its preeminence over problem solving, “The difference between a more and a less creative outcome

does not lie in the ability to solve the problem, but in the capacity to formulate it in an original way” (Csikszentmihalyi, 1988a, p. 185).

Clearly, problem finding and solving are strongly intertwined, resulting in meaningful problem-solution pairings. “The way in which the problem is conceived decides what specific suggestions are entertained and which are dismissed; what data are selected and which rejected; it is the criterion for relevancy and irrelevancy of hypotheses and conceptual structures” (Dewey, 1938, p. 112). Yet, Simon (1988) and Csikszentmihalyi (1988a) fiercely debated the respective nature of problem finding and solving. The former argued that problem finding, similarly to solving, is reducible to rational heuristics and can be replicated by computer models (Simon, 1988). The latter contended that problem finding is influenced not just by a rational logic but also by affective and motivational factors that can hardly be automated (Csikszentmihalyi, 1988b), hence requiring a different approach than problem solving.

In the following, we review the main insights from academic debates on problem finding across various disciplines. Indeed, the dominant view of problem finding is the discovery of unknown problems or unknown aspects at the core of known problems. These studies explicitly or implicitly draw on Simon’s science of design (Simon, 1973). We argue for more scholarly attention to designing novel, meaningful problems through reframing, an approach that builds on Donald Schön’s (1979) view of design as reflective practice inspired by John Dewey’s (1910, 1920) ideas.

2.1. Psychological perspectives

Psychology offers abundant evidence on problem finding (for a recent meta-analytic review, see Abdulla et al., 2020). Psychology scholars, especially those studying creativity, have analyzed the cognitive and behavioral drivers of problem finding and its effects at the individual level. A positive association between problem finding and innovation performance (measured as solution originality or quality) has been consistently found in studies on children (Okuda et al., 1991), undergraduates (Reiter-Palmon et al., 1997), and professionals (Basadur et al., 1982), the latter showing that engineers are more familiar with problem solving than finding, thus requiring longer and more intensive training.

In terms of the drivers of problem finding, Csikszentmihályi and Getzels (1971) find that art students enacting problem discovery-oriented behaviors generate more creative works than those jumping into solutions. Discovery-oriented behaviors include exploring more aspects of the situation through multiple sensory channels before defining and framing the problem based on unusual elements. Recent studies introduce systematic constructs to measure these behaviors, such as creative science problem finding (CSPF) and creative problem finding ability (CFPA) (Han et al., 2013; Hu et al., 2010). An individual’s ability to formulate multifaceted and purposeful questions is a core element of both.

The notion that working on the formulation of a problem can be key to its subsequent solution is central in Gestalt theory, a particular branch of psychology. Gestalt scholars discuss how successful solutions to insight problems (i.e., problems with a structure and elements that can be manipulated) are achieved through a shift in how the problems are viewed (restructuring) (Ohlsson, 1992; ter Hark, 2010).

Generally, psychology studies see creative problem finding as the result of superior individual cognition. By inquiring into existing contexts and experiences, innovators discover the nature of the problem and aspects thereof that they were unaware of, in turn suggesting effective solution strategies.

2.2. Organizational perspectives

Management studies extensively adopt the problem solving perspective (Hsieh et al., 2007; Nickerson et al., 2012), yet contributions to research on problem finding are limited and scattered across disciplines,

² Boland et al. (2008) define functionality broadly as the betterment of the human condition, arguing that it can never be fully realized.

including strategy, innovation, and entrepreneurship. The focus of these studies is on organizational settings, also relevant in this paper.

In strategy, influential scholars have indicated problem finding, framing, and formulating as critical to long-term business performance, since “these activities influence whether or not, in what direction, and for whom an organization creates new value” (Nickerson et al., 2012, p. 53). Given the complex and ill-structured nature of problems, it reduces the risk of solving the “wrong” problems.

Baer et al. (2013) investigate the causes of narrow problem formulations that constrain the generation of value-creating solutions. These include tunnel vision, the propensity to consider only information commonly held among individuals at the expense of uncommon information, the inclination to prematurely foreclose the formulation of the problem without exploring alternatives. They then propose a “collaborative structure inquiry” process that mitigates these impediments and enables more comprehensive problem formulation. Csaszar and Levinthal (2016) take a contingency view of problem finding and show that the value of searching for alternative problem representations increases with the problem’s complexity but decreases with severe time constraints. Von Hippel and von Krogh (2016) report cases of problems and solutions discovered together in a simultaneous and serendipitous way, contrary to the common expectation of problem finding as a deliberate act preceding solving. These “need-solution pairs” emerge when innovators encounter a solution worth implementing in their own context, concurrently recognizing the existence of a problem worth solving that they were neither aware of nor had searched for. To favor these discoveries, the authors suggest innovators tap into external problem solvers who bring novel views and distant knowledge.

In innovation, the growing popularity of design thinking as a creative method for new product development has brought greater attention to problem finding (Carlgren et al., 2016). Mainstream design thinking process models (e.g., the double diamond and Stanford d.school’s five hexagons) include dedicated problem exploration and definition steps (Liedtka, 2015). In these models, divergent thinking is not limited to brainstorming solution ideas but includes the preceding “painstorming” of problem symptoms and causes. The goal is to question the initial formulation of the problem and find a better one (Micheli et al., 2019).

Implicit in design thinking studies is the notion that the wicked and systemic nature of many problems today implies the existence of aspects of the problem that are not known by the innovator. One reason for these knowledge gaps vis-à-vis the surrounding environment is that relevant user needs can be hidden (Goffin et al., 2012). These are demands that users have but have not consciously recognized yet or are unable to articulate, a phenomenon called the say/do gap (Liedtka, 2015). For these latent needs to emerge, design thinking advocates immersion in the natural habitat and lived experience of users, observing their behaviors first hand (Brown, 2008). These methods help innovators develop empathy for those they want to innovate for (Kelley & Kelley, 2013).³ Another principle is the focus on extreme users, since unaddressed needs are more salient and thus easier to detect among the fringes of the user population, the “outliers” whose demands are amplified or perceived ahead of the rest of the population (Brown & Katz, 2011).

A second reason for the innovator’s knowledge gaps is that in complex systems, key elements like root causes lie beneath the surface of symptoms, pain points, and other observables (Furr & Dyer, 2014). Understanding the structure of the problematic situation requires a significant analytical effort. Why-questioning techniques, such as the five-whys used at Toyota, result in tree diagrams that decompose the problem into multiple levels and spotlight cause-effect relationships (von Hippel & von Krogh, 2016). Frishammar et al. (2016) show that

setting problem boundaries and mapping connections from symptoms to root causes are critical to the development of radical innovations.

Overall, the management studies reviewed see problem finding primarily as a discovery act. User research and deep-dive analyses help the innovator build a map of the problem and comprehend its precise nature. Just like a treasure hunter, the innovator seeks a path to the root cause(s) or area(s) deemed most relevant, thus constituting the best angle to tackle the problem from. This discovery represents the so-called moment of clarity (Madsbjerg & Rasmussen, 2014) when the “correct” formulation of the problem is found and drives the generation of high-value solutions (Wedell-Wedellsborg, 2017). The case of Rivigo, an Indian unicorn in the trucking industry, illustrates this approach (Gupta et al., 2020). Its founder, Deepak Garg, set out to address the systemic lack of truck drivers in India and managed to find its key root cause: the fact that truck drivers are constantly on the road, away from home for long stretches. The solution he developed, a system inspired by relay races, was based on building frequent pit stops into the routes for drivers to exchange trucks and itineraries, thereby attracting new drivers and creating enormous value for logistics companies.

Differently from the above, there are some management approaches that do not center on investigating existing user needs and past causes to find problems. Instead, they “invent” problems by envisioning what could provide new meaningful value to stakeholders in a world that changes (Van der Bijl-Brouwer & Dorst, 2017). These future visions create a gap with current reality, and their realization require overcoming specific problems.⁴

In entrepreneurship, Felin and Zenger (2009) argue that success derives from developing new theories of value that make an entrepreneur see the world differently. New theories stem from the entrepreneur’s contrarian beliefs that question long-held assumptions in a domain. This was key in the entrepreneurial journeys of Sam Walton, Steve Jobs, and Howard Schulz. The authors propose the Value Lab approach (Felin et al., 2020) to support entrepreneurs in forward-looking theorizing about possibilities beyond extant realities, realizing these theories through iterative experiment-based problem solving.

Similarly, the blue ocean strategy (Kim & Mauborgne, 2014) challenges established paradigms in industries to invent new market spaces that did not exist, as Cirque du Soleil did by creatively merging circus and theater entertainment. The creation of blue oceans involves, amongst others, a redefinition of the strategic problems based on new or hitherto overlooked value parameters.

In innovation, Verganti (2008) introduced the concept of design-driven innovation, an approach aimed at radically changing the meaning of products and services through a deep understanding of broader changes in society, culture, and technology. The meaning of a product is captured by the utilitarian, emotional, and symbolic value it offers users. By focusing on meaning, Verganti emphasizes the perpetual human search for value and worth. People are not seen as “bags” of needs (negativities) that must be fulfilled, but as reservoirs of aspirations (positivities) that constantly evolve to maintain fit with the changing environment (Verganti, 2017). For example, Swatch proposed a new meaning for watches as a fashion accessory that can be frequently changed to match the person’s mood or colors of the day. A very different meaning to watches as jewelry and as instruments of time that emerged in the 80 s with the introduction of Quartz technology. Therefore, innovating the meaning implies generating novel problems worth addressing to offer the desired value.

2.3. Design perspectives

The design field provides a rich and insightful body of scholarship on problem finding. Christiaans and Restrepo (2001) distinguish be-

³ Empathy refers to a person’s ability to accurately recognize another person’s thoughts and feelings, and emotionally react in response (Atkins et al., 2016).

⁴ Unsurprisingly, the authors of these approaches do not explicitly position their works in the problem finding debate, yet offer valuable inputs to our conceptual development.

tween problem-oriented designers who formulate problems through abstract concepts and relations, and solution-oriented designers who create narrower problem formulations that already hint at physical objects. Casakin's (2012) experiment indicates that the quality of final solutions increases when the designer re-interprets a situation from different perspectives and uses metaphorical thinking to link the problem with remote domains. According to Cardoso et al. (2016), asking deep-reasoning and generative questions helps switch viewpoints to the problem.

The extensive work of Kees Dorst (2011, 2015) has profoundly contributed to problem finding research. According to Dorst, reframing is at the core of design practice. To generate innovative solutions (what), designers iteratively and abductively develop frames, i.e., combinations of desired outcomes (why) and working principles (how), creatively moving from "why" and "how" to "what" and back until a meaningful fit is achieved (Steen, 2013). In this view, design is the dynamic co-evolution of the problem space and the solution space (Maher et al., 1996).

The work of Dorst and other design scholars (e.g., Stomppff et al., 2016; Van Leeuwen et al., 2020) draws on Donald Schön's (1979) view of design as reflection in action. The designer is in a reflective conversation with the situation that is interpreted in a fluid, ever-evolving fashion, questioning the established patterns of relationships, and generating new ways of acting in the situation (Schön et al., 1996). This sensemaking effort is driven by the subjective lenses designers use to frame the problem in their mind.

It is somewhat surprising that the management literature building on design principles and practices (Micheli et al., 2019) has only picked up Schön's insights to a limited extent. A different school of thought, namely Simon's (1973) design as science, has had much greater influence. In Simon's work, design is seen as a search within a problem space that is objective and knowable. Boundedly rational designers should focus on finding an acceptable solution to ill-structured problems by narrowing down the problem space to constrain and direct the search effort. We believe that solely relying on Simon's ideas can limit our understanding of problem finding and its potential for innovation in organizational settings. We therefore draw on Schön's insights on framing and reframing to conceptualize problem finding as the design of more meaningful problems.

3. Key concepts of designing problems worth solving

The design of a more meaningful problem stems from the generation of alternative frames to look at the situation at hand and make sense of it in original ways (Dorst, 2011). The frames can be borrowed from a different domain or be completely new (Cukier et al., 2021). Either way, they "give new life" to the problem by suggesting other realms of the human experience not intuitively associated with it, and once added to the scope, can lead to groundbreaking solution possibilities (Boland et al., 2008).

How does an entrepreneur design a novel frame? As with any task in uncertain conditions, it begins with the divergent creation of multiple possible frames followed by the convergent selection of the one that makes most sense.

3.1. Generating novel problem frames

3.1.1. Problem expansion and elevation

Generating novel frames requires decontextualizing the problem from the situation in which it originated. Problem expansion and elevation are two complementary ways of doing so. The former consists in expanding the problem scope by adding elements that were not included in its initial characterization (Liedtka & Ogilvie, 2011; von Hippel & von Krogh, 2016). This can be done by considering moments and activities in the user experience that are outside the specific problem situation, albeit connected to it, and/or enlarging the list of actors taken into account beyond the inner circle of stakeholders evidently "at play" and

affected by the problem. "[Beyond] that core group is the wider field of parties who have not participated in the problem situation before but who have a potential influence, and could develop a direct interest in the problem if it were framed in a different way" (Dorst, 2015, p. 136).

Although Dorst's logic is partly circular – to reframe, the entrepreneur should consider a wider field of parties, but to understand which additional parties to consider, they should have a new frame already in mind – unexpected actor categories bring their own set of needs, aspirations, values, interests, priorities, and behaviors that can push the problem in a new direction. Speculating which new parties to include in the scope is a creative task, whereas identifying their needs, aspirations, and so forth, is mostly analytical.

To exemplify problem expansion, Dorst reports the case of the marathon in the Dutch city of Eindhoven, a sporting event that greatly inconvenienced its citizens due to road closures and traffic diversion. Originally framed as a traffic control problem, the solution came from considering a novel actor category unrelated to the problem as it was originally framed: high-tech companies, which are abundant in Eindhoven. The combination of two insights (i.e., that these companies constantly need new staff and that most runners are highly educated often with a STEM background), led designers to reframe the sporting event as showcasing the region of Eindhoven as a good place to work and live. This led to the solution of dividing the marathon circuit into themed zones where local companies and residents would organize activities to expose runners and their families to the possibility of a life in Eindhoven. Seen in this new light, the marathon provided an attractive business proposition to companies and made more sense to residents, as it was now aligned with the city's primary needs and even alleviated traffic issues by drawing crowds away from the city center into the suburban areas where the companies were located.

To note, the problem expansion that is instrumental to reframing is at odds with problem decomposition, a key principle of Simon's (1973) science of design on which problem-finding-as-discovery approaches build. Designers following the decomposition logic tackle complex, ill-structured problems in an "immediate problem space", a portion of the whole space that instead is too large and ill-defined to analyze. In other words, they carve off a relevant piece of the problem and find a logical and feasible solution to it. Churchman (1967) argues that this tactic can only tame the growl of the wicked problem, but not avoid its bite! Conversely, the sensemaking process for designing novel problems draws on the epistemologically unconfined exploration of new conceptual and cultural territories typical of designers (Cross, 1999; Michlewski, 2008).

A second way to decontextualize the problem is through elevation, a figurative term for the cognitive abstraction of the problem (Beckman & Barry, 2007).⁵ This helps recognize the universal human values that overhang a situation. Scholars have labeled them with different but overlapping terms⁶: job-to-be-done (Christensen et al., 2016), the ultimate reason why people "hire" a product or service in a specific circumstance; meaning (Verganti, 2017), the special importance and purpose of something that makes individuals feel their life is valuable; and higher-order themes (Dorst, 2015) that are relevant to the problem situation.

A problem can be elevated through reiterated why-questioning: "why does the problem matter?" followed by "why does the reason you've just given, matter?". Theodor Levitt's famous quote "people don't want a drill, they want a quarter inch hole" illustrates this technique: why does making a hole matter? Because (certain) people want to hang pictures on the wall. Why does this matter? Because they want to preserve memories and share them with family and friends. And so forth.

⁵ Whereas some scholars see this abstraction process as digging deep into profound human meanings (metaphorically, a descent), we see it as an ascension to higher-order goals. We use the term "elevation" to also differentiate it from the deep-dive analysis of root causes.

⁶ In this paper, we generally refer to them as universals.

This sensemaking effort creates a ladder of connected, increasingly abstract meanings on top of which “sit” universal themes (e.g., connection, memory, belonging) that although independent of the initial problem context, describe what the situation down there “is all about” (Van der Bijl-Brouwer & Dorst, 2017).

To note, the nature of the “why question” posed here differs from that used for a root cause analysis. This distinction, often overlooked by scholars (e.g., Furr & Dyer, 2014), is important. The question “why does the problem occur?” in the five-whys technique allows tracing the origins or sources of the problem (von Hippel & von Krogh, 2016). Conversely, repeatedly asking “why does it matter?” explores the relevance of the problem in the life world of people (van der Bijl-Brouwer & Dorst, 2017). The resulting richer understanding of higher-order human values prepares the ground for valuable reframing.

While elevating the problem to spotlight the overriding “why(s)” is a prerequisite to designing novel problems, it is not sufficient. Take the case of the pharma company aiming to innovate its smoking cessation offering (Liedtka & Ogilvie, 2011). Through field user research and pattern recognition, they discovered that youngsters see their smoking habit as part of a temporary lifestyle choice. Based on this insight, differing from the common expectation of smoking related to a chemical addiction to nicotine, the company developed solutions including coaching programs and social support services to help young smokers choose an alternative lifestyle. These created a new growth platform for the company and differentiated it from its competitors.

In this example, discovering the “why” through a fuller understanding of human needs was a pivotal moment because it oriented the search towards solutions that better addressed what ultimately mattered to users. However, this “why” was not the innovator’s original creation, as it had characterized the problem from the start and already drove user choices, even if the innovator was mostly unaware of it.⁷ Designing novel problems worth solving occurs when the entrepreneur innovates the “why” and not just the “what” (solution).

3.1.2. Speculative leaps

More than the analysis of the current reality, designing novel problems requires speculative leaps to imagine which realms of human experience can be added to expand the functionality of a domain (Boland et al., 2008). In Verganti’s (2008) words, which other utilitarian, symbolic, emotional dimensions could characterize the new meaning? Or drawing on Dorst’s (2015) work, which alternative higher-order themes, once marginal or non-existent, could matter in the situation? Despite the different vocabulary, these constructivist acts correspond to building and fitting an original frame onto the problem through a generative combination of problem expansion and elevation. Figuratively, these speculative leaps are like jumping off a ladder of meanings made of answers to reiterate the questions of “why does the problem matter?” (Fig. 1). While vertically ascending the ladder means eventually reaching universals that characterize the current problem frames, a swerve to the side allows exploring alternative universals as entrepreneur ask themselves “why else could it matter?” or “what is another potentially important reason why this problem will matter?”, immediately reflecting on whether these reasons and the frames they suggest make any sense.

Foreign or peripheral universals can become the new core in a situation and suggest new frames through different mechanisms. First, as a consequence of the constant evolution, and at times revolution, of the social and technological environment. If considering photos, the advent of digital photography led the once-dominant frame of memory preservation to become secondary, replaced by the frames of real-time communication and self-expression. The success of photo-sharing social

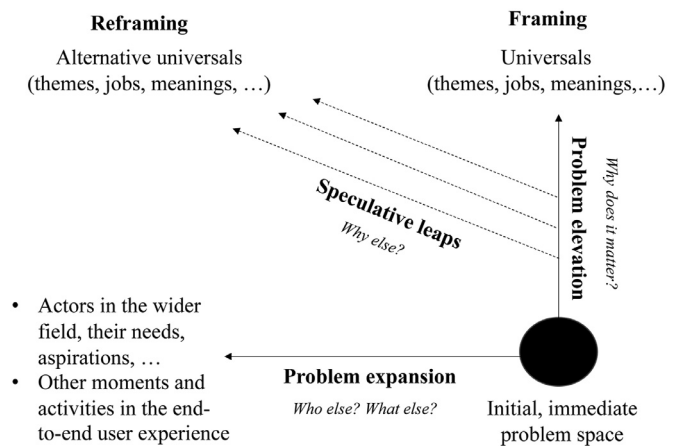


Fig. 1. Problem expansion, elevation, and speculative leaps for designing novel problems.

networks like Instagram signals today’s prevalence of these frames. Although Kodak invested early and extensively in digital technology, it neglected to see that digitalization was causing a paradigmatic shift in the meaning of photos. The fact that Kodak was fixated on the traditional view of photos as memory preservation artifacts is indicated by some services they introduced: booths for printing digital photos and print-one-get-one-free offerings.

A second mechanism is analogical thinking. When the problem space is opened up and elevated to an abstract level, it is easier to identify isomorphic relationships between domains (Simon & Hayes, 1976) to create connections between actors sharing similar universals and across areas of human life, brokering perspectives from a distant source domain to look at the target problem in a new light (Pee et al., 2015). These non-trivial associations are constructed in the entrepreneur’s mind rather than seen. It is a design act that transforms the abstraction ladder into a ramified and potentially unbounded tree of meanings.

Prodigy Finance (Kim & Mauborgne, 2019) founded by INSEAD MBA alumnus Cameron Stevens in 2007 is a case in point. The problem that this start-up set out to address was that many talented youngsters from emerging countries lacked the money to finance their MBA studies in business schools in developed countries. The solution that Prodigy Finance first developed, a predictive lending model for banks, did not work because it ran counter to the traditional credit worthiness criteria that banks used: loan recipients must be local residents with collateral and a past credit history in the host country.

Key to breaking the gridlock was abandoning the conventional economic frame (how might we ensure that lending to international students provides steady financial returns to investors?), and designing an alternative, social, and emotional frame: how might lenders exert positive social impact in the developing world while receiving emotional gratification from doing so? Inspired by this new frame, Prodigy Finance bypassed banks as a funding source and created a peer-to-peer model where the lenders were empathetic alumni from the same school as the MBA applicant who cared about establishing a cross-generational emotional connection to future leaders and close the skills gap between countries. These meanings, historically marginal or non-existent in the financial domain, were thus designed by the entrepreneur. An inspiring analogy is the long distance adoption of children, a diffused practice in the charity domain. The peer-to-peer model later evolved into a larger scale community model for wealthy individuals who could invest based on affinity and the desired social impact. As of 2022, Prodigy Finance has lent over 1 billion dollars to more than 20,000 students with less than a 1% default rate.

Not only is the abductive sensemaking process for reframing open-ended, but it is also prospective (Morais-Storz et al., 2020; Stigliani & Ravasi, 2012). It looks at the future of the problem: the entrepreneur

⁷ Universals are abstract concepts that are seldom discussed in normal parlance or thought about in the use situation. Detecting them requires intense sensemaking.

explores the possibilities of the problem beyond extant realities, aspired possibilities sparked from aligning environmental changes with personal purpose (see next section). Problem designers act in the realm of speculative thought for a larger portion of the process than discoverers, in line with design practice as “a prepositional and creative discipline [that] always sees things in the future and is moving towards them” (Michelski, 2008, p. 384). Borrowing a medical term, problem finding as design concerns the prognosis of the problem, its future development, rather than the retrospective diagnosis prevailing in problem finding as discovery.

We have thus far associated the design of novel problems with reframing. An interesting question is: can reframing also occur with a problem discovery approach? We argue that it is possible, but that the retrospective analysis and problem decomposition typical in problem discovery do not facilitate reframing. The identification of a critical root cause invites reformulating the problem around it, but reformulating the problem is not necessarily reframing. A change in frames requires first identifying the overarching paradigm through elevation and then “escaping” it by fundamentally revisiting the assumptions on which it is built. The multiple case study of Morais-Storz et al. (2020)) shows that engrained assumptions go unquestioned when teams make sense of a problem only retrospectively. Digging deep into the root causes may even reinforce the old frame, since the entrepreneur develops valuable knowledge of the problem’s origins and might thus have a harder time abandoning the frame, even if a significant environmental shift has made it obsolete. This phenomenon is called fixation (Purcell & Gero, 1996).

3.2. Selecting a novel frame

Among the novel frames being designed, which one is eventually chosen for the actual reframing that takes the problem in a new direction? We posit that convergence is a matter of how meaningful the frame, and hence the resulting problem, is perceived by the entrepreneur, and that higher meaningfulness vis-à-vis other frames, including the old one, is a matter of fit between the entrepreneur’s purpose, values, and the emerging macro-environmental dynamics (Lai & Grønhaug, 1994).

Worth noting is that when designing a new problem, there is no presumption that the outcome will be the “right” problem. While a problem discoverer might claim this when drawing a path to the core root cause or ultimate job, it would be naïve to believe that the entrepreneur will design the one right problem, given the speculative, aspirational orientation of the task. The frame being created is one among several possible directions the problem can take in the future. What matters is that the new problem is perceived worthier than before and in line with the impact the entrepreneur seeks in view of where the world is heading.

3.2.1. Psychological factors

Internally, reframing has a significant inside-out connotation: the design of a problem worth solving can be triggered by personal malaise with the status quo, a perception that the givens of the industry no longer fit the shifting context, the intention to drive transformation in a certain direction. Prodigy Finance’s founder, South African Cameron Stevens, experienced first-hand the problem he was determined to solve as he had to postpone the start of his own MBA due to being denied a loan by banks (Kim et al., 2019). He grew frustrated with the “orthodoxies” of the lending sector that made little sense in the emerging context: why is lending based on past records when modern software technology allows accurately predicting future income? Why must access to finance be local when talent moves globally? Why does no one want to invest in talented people already admitted to top business schools with the evident ability and willingness to pay back in the future? Stevens’ strong belief that talented youngsters should be given equal op-

portunities wherever they come from kept him going despite the early setbacks.

The design perspective to problem finding emphasizes the central role of personal worldviews (*Weltanschauung*), no matter how subjective and biased, as long as they lead to frames that are original to the problem at hand and aligned with the way the world is changing. For instance, the reframing of mobile phones from a means of voice communication to a device remotely controlling our lives was driven by visionary Apple designers under the leadership of Steve Jobs largely embodying Apple’s identity and purpose. These aligned with ongoing developments in electronic technologies and broadband cellular networks with the emergence of platform models, resulting in a set of novel problems to address in order to create a more meaningful and modern experience for users. Illies and Reiter-Palmon (2008) find that leaders construct problems in a way that fits their value system. In the context of social movements, frames are shown to directly connect with the personal and social identities of activists (Creed et al., 2010).

In many reframing cases, leaders embody the frame pursued and project a strong emotional conviction about why it makes more sense. In an interview, Elon Musk stated: “I was very curious about what’s the meaning of life. I had a kind of existential crisis when I was 11 or 12 years old, came to the conclusion that we do not know the answer, but if we increase the scope and scale of civilization, we will have a better chance of understanding the meaning of life. Therefore, we should strive to expand the abilities of our consciousness to better understand the questions that we have to ask ourselves for which the answer is the universe” (Musk, 2021, 1:01:18). To increase the scope and scale of multiplanetary civilization, dramatically reducing the cost of space launches and expanding the number of opportunities is critical (Kobylnski, 2022). For Musk and the company he founded, SpaceX, achieving this revolves around the reusability frame, a paradigm shift vs. the government age of space, where making booster rockets fully reusable is the very meaningful problem to solve (Cukier et al., 2021).

On the one hand, the inside-out connotation of reframing contrasts with the invitation for deframing from the user-centered design thinking literature (Liedtka, 2015). These works recommend collecting and analyzing data in the most unfiltered and unbiased way possible with a beginner’s mindset free from preconceptions (Madsbjerg & Rasmussen, 2014). For entrepreneurs seeking to discover the “true” essence of the problem, they suggest quieting down the lenses through which they perceive reality, and avoiding putting too much of themselves into the problem. This advice is based on the limitations of frames as incomplete and biased representations of reality that can lead to errors in judgment (Hong & Page, 2009).

On the other hand, inside-out reframing highlights the inherent similarity between design and leadership (Boland et al., 2008). Every designer is a leader in the truest and fullest sense of the word, because every design process is about intentionally leading the world into a new reality worthy of use as human beings (Nelson & Stolterman, 2014). This future hinted at by the new frame is however far from reality. Mike Maples of Floodgate, a venture capital firm, observes: “Startups are not about mapping the territory or the market. They are about finding something valuable about the future and then starting a movement of people who believe what you believe until everybody believes” (Wuebker et al., 2021, p. 39). A complex endeavor given the contrarian nature of reframing (Felin et al., 2020). Yet, the new problem should make sense to the entrepreneur before it does to anyone else.

3.2.2. Environmental factors

Drawing on personal views to reframe problems does not mean downplaying the importance of the external environment. On this aspect, we see an important difference between problem discovery and problem design. In the former, the environment that is in focus is the narrower, immediate use context, whereas macro-contextual forces tend to be assumed away and treated as constants, because they are seen as out-

side the scope of the intervention.⁸ In the latter approach, conversely, the emphasis is on the wider social, economic, cultural, and technological environment (Ramirez & Wilkinson, 2016). Socioeconomic changes, cultural dynamics, and the new possibilities afforded by technological progress are treated as degrees of freedom that inspire the generation of new frames and the assessment of their meaningfulness. In the case of Prodigy Finance, the new social and emotional frame designed by the start-up aligned with the growing role and activity of impact investors whose mission is also to exert a positive impact on people's lives and society at large. This trend continues today, as more financial players recognize the responsibility to help society tackle the grand challenges it faces.

Foresight skills are key to the problem-designer entrepreneur who envisions how ongoing or future environmental forces (depending on how far ahead in time the design is projected) will influence what is meaningful to humans. Ten or twenty years from now, the circumstances in most domains will be significantly different, as will what matters to people in their lives. An emerging technology, even originating from a distant domain, can drive the fundamental reframing of the meaning of a situation or product. Verganti (2011) calls this phenomenon a “technology epiphany”.

Take blockchain, a rapidly-diffusing distributed ledger technology with supposedly game-changing potential. By recording transactions among multiple parties in a verifiable, immutable, and transparent way, blockchain offers for the first time a method for attaining scarcity in a digital environment (O'Dwyer, 2020). Digital assets can thus be perceived valuable in the same way as physical assets for their physically imposed limitations. This can have revolutionary implications in the art domain (Kaczynski & Kominers, 2021): blockchain-based NFTs (non-fungible tokens) can radically change the meaning of artworks, challenging the traditional “ownership = possession” equation and privileging the former over the latter. An entrepreneur having an epiphany envisions new experiences that the technology can give rise to, and based on these, designs novel problems to be solved and the set of artifacts to do so.

However, focusing on the narrow, immediate use context and employing user research techniques, as in problem discovery, is unlikely to generate epiphanies and reframes. User insights are rooted in the user's situation at that moment. “[Users] tend to reason from what is instead of what could be; their frame of reference is the present, not some future world” (Hekkert & van Dijk, 2011, p. 184). Verganti (2008) argues that because users are immersed in the current sociocultural context, they are unlikely to appreciate interpretations that are not in line with today's zeitgeist. The implications are twofold: first, user data from interviews and surveys will likely suggest that new frames are delusional or make little sense and easily dismiss them, thus not helping establish the fruitfulness of new frames; second, the entrepreneur's aspirational and contrarian beliefs are key to radical reframing (Felin et al., 2020).

4. Designing problems worth solving in practice

Reframing is often presented as a mysterious leap, largely dependent on the designer's skills and inspirations. However, concrete methods to reframe problems exist. While offering a systematic typology is outside the scope of this paper, we here classify the methods according to the key driver supporting reframing (see Fig. 2). In frame innovation (Dorst, 2015), reframing occurs through making sense of the problem, in scenario thinking (e.g., Ramirez & Wilson, 2016) and vision in design (Hekkert & Van Dijk, 2011) through making sense of the wider context, in innovation of meaning (Verganti, 2017) through making sense of the

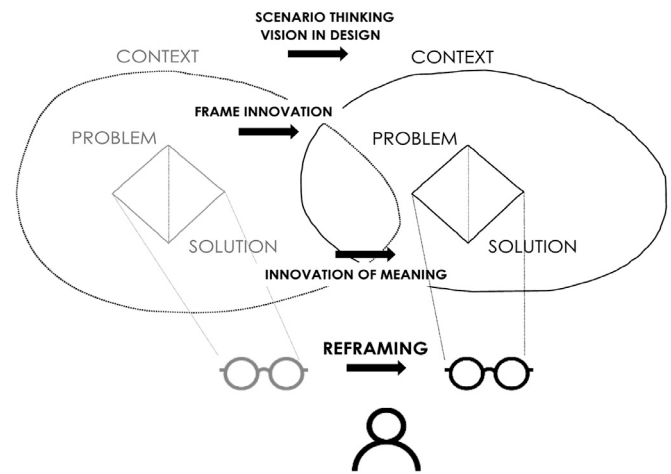


Fig. 2. Methods according to the reframing driver.

solutions. These methods variously support the addition of novel, previously lacking elements and perspectives to the problem scope, be they actor categories and their needs, themes, meanings, from other domains, and/or from the future. This addition to the “what is” of the situation that aims to expand its functionality (Boland et al., 2008) has a prospective, constructivist nature, making this kind of problem finding a design (vs. discovery) act. In the following, we introduce these methods and their main logic.

4.1. Reframing through the problem: Dorst's frame innovation

Dorst's frame innovation method is a nine-step process (for a complete description, see Dorst, 2015). Key to reframing are three central steps, namely field, themes, and frames. Reframing is achieved through reflecting on and speculating about the problematic situation that triggered the process. Although most published evidence of the use of frame innovation concerns social problems in the public domain, the method is also useful in corporate and entrepreneurial contexts (Dorst, 2015).

Following the expansion of the problem space through imagining new possible actors (field phase) (see Section 3.1), all the identified needs, aspirations, interests, independently of the originating actor, are clustered into coherent groups according to an overarching theme (e.g., loneliness, dependence, identity formation) that ties the elements together. This technique is called affinity diagramming (Beckman, 2020). Due to the preceding problem expansion, new themes can emerge beyond those intuitively stemming from the initial problem setting.

Themes are then explored in earnest and at length (themes phase). Hermeneutic phenomenology is the social science method Dorst suggests to thoughtfully discern and investigate the meaning of each theme in the wider scheme of things (Van Manen, 1990). Van Leeuwen et al. (2020) list five specific techniques to create rich accounts of the phenomenology of a theme. Two of these, the stakeholder and researcher perspective, are based on personal experiences. They consist of asking an actor (stakeholder) or a designer (researcher) to recount relevant emotions connected to a theme in other situations than the problem's. Two techniques are analytical: the scientific (philosophical) perspective consists in reviewing scientific literature (philosophical treatises) about the theme to learn about its different manifestations, drivers, and consequences. In the fifth evocative perspective, designers get inspiration from how various forms of art and culture (paintings, poetry, music, etc.) interpret and express the meaning of a theme. The outcome of the phenomenological investigation is a nomological network for each theme, essentially a wordcloud that represent the related psychological, social, cultural and technical nuances connected to each other and to the central theme.

⁸ Dorst (2015) calls “freeze the world syndrome” the tendency to isolate the problem as if the world were stopping despite that the “rules of the game” keep changing all the time. Undesired results of this tendency are static maps of the problem with short expiration dates, as they become irrelevant when the macro-environment shifts significantly.

In the frames phase, the nomological networks act as springboards for the creation of novel frames. Through analogical thinking (Casakin, 2012), the designer interrogates how the elements of a theme are dealt with in other domains outside of the problem context. In what other sector or area of human life are these aspects realized at an advanced level? Distant domains offer access to a large body of knowledge, experiences, patterns of relationships that can form the basis of new frames. In frame innovation, frames are expressed through metaphors (“if the problem situation is approached as if it were..., then...”), since they embody sharp patterns of relationships and evoke a clear picture in the mind of all stakeholders (Pee et al., 2015).

Van der Bijl-Brouwer and Dorst (2017) describe an innovation project aimed at offering cover from bad weather to children when riding a carrier bike (a bicycle with a box in the front). Beyond intuitive themes like protection, safety, comfort, the team broadened and elevated their reflection to the general theme of parenting. Exploring the phenomenology of what it means to be a parent suggested a novel relevant theme, guidance, in which the child is guided by the parent as they interact with the world. In a bike ride around the city, parents talk to their children, point out interesting things, explain and tell stories about these. Drawing on a nuanced nomological description of guidance (for example, the stimulation of the child’s observation of the surroundings, the provision of information by the parent, and the child’s subsequent reaction of wonder and surprise), the team formed a new frame through the metaphor of a bike ride as if it were a joint exploration of the world, which led the project toward a significantly different trajectory from that suggested by the initial protection frame.

4.2. Reframing through the context: scenario thinking and vision in design

Scenario thinking consists of envisioning plausible worlds in a distant future, typically 10 to 20 years ahead (Garvin & Levesque, 2006). Starting from a focal issue, another term for the initial problem or opportunity, entrepreneurs build scenarios by combining macro-environmental forces, such as geopolitical trends, regulations, economic, societal, and technological dynamics (Ramirez et al., 2017). Scenarios act as reframing devices for different reasons (Ramirez & Wilkinson, 2016).

First, focusing on the outer environment beyond the organization’s direct influence pushes entrepreneurs to explicitly consider factors that are distant from the focal problem and the conventional frames in an industry. Entrepreneurs then make prospective sense of the implications of the factors, generating multiple imagined future trajectories. Dunne and Raby’s (2013) future cones visually represents the multidirectional speculative leaps triggered by scenario thinking.

Second, scenario thinking entails the generation of alternative scenarios, not forecasting one single estimate. The scenarios are by design⁹ orthogonally different from each other, as they result from the two-by-two crossing of opposite outcomes of the driving forces (Garvin & Levesque, 2006). Generating scenarios stretches entrepreneurs’ thinking as they must actively consider events they deem as remote possibilities, and entertain the possibility of futures that are very different from today, from each other, and from the so-called ghost scenario. The ghost scenario is the expected future obtained by linearly projecting trends ahead in time based on established beliefs and frames.

Once imagined, scenarios are brought into life as vivid stories that provide an immersive description of a holistic world resulting from the complex interaction of several factors. The zoom then moves from all-round worlds into the human experience as the entrepreneur prospectively makes sense of what will be meaningful to people should each newly created world materialize. What will they come to value? What markets might emerge because of this? The outcomes of this “learning from the future” reflection are imaginative frames situated in the future

to look at the focal issue (Ramirez & Selin, 2014). In the final backcasting step, the entrepreneur travels back to the present, using the frames “from the future” to re-perceive the present based on different beliefs and aware of fresh strategic possibilities (Ramirez & Wilkinson, 2016). What actions today can lead to a desirable future? What solutions can be developed? In Von der Gracht and Stillings (2013), a global producer of high-performance materials exemplifies this approach.

A design method based on scenario thinking is vision in design (VIP) (Hekkert & van Dijk, 2011). In the first creative step, the designer builds future contexts by considering both the principles and the states, which are time-invariant “laws” of nature and human psychology, as well as the developments and trends, which are changing patterns. The main goal is to shift the view of what a phenomenon, an area of human life, such as “work”, “love”, “family”, might be years ahead. The imagined future context is then used as a frame of reference by the designer who reflects on its implications for human behaviors and experiences, and in turn on the innovative products and services that can afford these experiences (Lloyd et al., 2006). While nonsensical today, as they might not solve any problems that people currently have, the product outputs of VIP are prospectively meaningful, as they fit the hypothetical conditions of the new context.

For instance, the emergence of the metaverse and decentralized autonomous organizations (DAOs)¹⁰ could lead to a paradigm shift in the role of work in people’s lives. Consider a future where work is not a nine to five activity within a stratified hierarchy in exchange for a monthly salary. Instead, new immersive virtual technologies and social arrangements allow extreme flexibility as to when, where, and how to work, the recording of each single piece of work on the blockchain rewarded with project-specific tokens, the dominance of “x-to-earn”¹¹ models over the traditional work-to-earn. In this world, the jobs-to-be-done by services once run by HR departments change fundamentally. Staff satisfaction will no longer stem from addressing problems such as career advancement, job security, and workplace conditions, but from novel problems worth solving. For instance, the deep financialization of the everyday (Wilser, 2022) associated with the tokenization of any asset, or the development of a sense of purpose as work commitments are fractionalized to the most elementary level, indistinguishable from personal life. Having created a vivid image of the future world, even if it does not yet exist, the designer can envisage what is appropriate for that world, which is unlikely to be a better functioning HR department.

To note, there is a key difference between scenario thinking in management practice (Garvin & Levesque, 2006) and VIP. The former advises managers to neither judge scenarios as desirable or undesirable nor develop a preference for one over others. The goal of this approach is to prepare for possible futures in volatile and unpredictable contexts, which is better achieved if managers have a broad, neutral view of the possibilities, instead of focusing on one desirable direction. Conversely, VIP expects designers to take a stand and stay true to themselves when choosing the factors to build a future context. This approach results in a worldview that, albeit capturing the relevant macro-dynamics, is personal and distinctive, and so is the vision statement describing how they intend to respond to this worldview (Van der Bijl-Brouwer & Dorst, 2017). The designer’s aim is not seeing the future to prepare for different circumstances but “seeding” the future by influencing the course of action (Wilkinson et al., 2013). In VIP, being assertive is more important than being comprehensive.

4.3. Reframing through solutions: innovation of meaning

In Verganti’s (2017) innovation of meaning (IOM) method, the design of a new frame is driven by reflections on the meaning of to-be

⁹ Assuming the most diffused tool for scenario generation, the 2x2 scenario grid, is used (Garvin and Levesque, 2006).

¹⁰ DAOs are digital-native communities functioning through smart contracts and organized through collective ownership and interest.

¹¹ The “x” in “x-to-earn” models can refer to engaging in sustainable behaviors, supporting other members, curating content, playing, learning, exercising, etc.

vis-à-vis existing solution. In 2011, Nest Labs innovated the meaning of residential thermostats. Historically, thermostats were intended to allow people to control the temperature in their homes. New technologies, such as programmable settings and touchscreen displays, have been introduced over time to perform this function better. Nest Labs took another direction: aware that family life is becoming more complex and unpredictable to regularly program a thermostat, Nest Labs leveraged technological advances in smart phones, sensors, and machine learning to create a thermostat that learns the habits of the household and takes care of the temperature settings itself. This new meaning for thermostats, from controlling the temperature to letting people forget about it and enjoy their time at home, reframed the problem worth solving for Nest Labs, namely higher automation and artificial intelligence.

Differently from most innovation methods, IOM invites entrepreneurs to immediately ideate new solutions to the focal challenge. This is typically discouraged by the user-centered design thinking literature as it can lead to the premature closure of problem exploration, fixation, and incrementalism. These fallacies are theoretically avoided in IOM because jumping to solutions is followed by jumping back to the problem to explore new frames. Therefore, solution attempts are a means to finding new meanings, and not to solve the problem. After having tentatively proposed a new solution, the entrepreneur reflects on what utilitarian, emotional, symbolic value it would provide to people and how this differs from existing offerings. To support the comparison of different whys, Verganti suggests using simple forms containing arrows indicating a change from an existing dimension of value to a proposed one (Verganti, 2017).¹²

More than providing a toolkit, IOM's core contribution to reframing practice is to propose an actor-oriented dialogic process through which the entrepreneur entertains conversations on the meaning with a growing number of diverse actors. The first step is individual, as participants articulate their own vision for the focal product based on the way they see the outer environment evolving and on their personal values. In the second step, pairs are formed where each innovator subjects their vision to the criticism of a trusted peer acting as a sparring partner, and vice versa. "Working in pairs creates an environment of instrumental intimacy in which people can sympathetically and constructively criticize each other" (Verganti, 2015, p. 4). The pair strengthens and synthesizes their differing perspectives into a bolder, more meaningful vision, which is immediately embodied in a concrete artefact. Without doing so, the proposal for the new meaning would be too abstract to properly communicate and be grasped by the rest of the team. In the third step, two or three pairs form a circle each that continues with the process of clashing and fusing visions to develop a stronger one. In the IOM process, the meaning and solution develop hand-in-hand in high-frequency iterations where reflecting on a new solution leads to new meaning, in turn leading to a new solution.

The final step consists in opening the conversation further to external interpreters, i.e., experts from far-flung fields who bring novel perspectives to the matter at hand (Verganti, 2011). These are professionals who are interested in similar phenomena that they address from different angles as a result of their affiliation, professional background, worldview. Interpreters challenge the meaning proposals developed by the circles and raise questions. They act as bridges between sociocultural worlds, allowing the transfer of frames across domains and their original recombination. For example, Philips tapped into interpreters to address the problem of child anxiety when undergoing medical scans. The team at Philips met with architects, interior designers, psychologists, developers of interactive software and hardware, light technologists, artists. This interaction exposed the team to different perspectives on what being

examined means. Insights from one interpreter, a clinical psychologist studying how pain affects children during treatment, suggested that a narrow focus on the diagnostic machine in the examination room, which was common in the industry, missed the fact that anxiety starts immediately as the child arrives at the medical center, mounts in the waiting room, and does not disappear after the examination. By expanding the problem scope, Philips designed a new user experience, named Ambient Experience for Healthcare, which combines ambient technologies, such as lighting, sound, movie projection, to spark relaxation throughout the child's end-to-end journey. This was a major strategic departure from the emphasis on diagnostic quality and speed of most competitors.

5. Discussion and conclusion

This paper introduces key concepts and practices to design novel, more meaningful problems, an alternative perspective to problem finding as discovery, which is mainstream in the management and entrepreneurship literature (Hsieh et al., 2007).¹³ In our conceptualization, a problem is designed (vs. discovered) when novel aspects (e.g., actors, meanings, themes) that were not part of the initial problem scope are added, offering a new view of the situation that allows expanding the functionality of the domain. This addition to the "what is" is constructivist and characterizes this type of problem finding as a design act. Designing a problem is mainly based on inventively reframing a situation through its expansion, elevation, and prospective sensemaking (Lai & Grønhaug, 1994; Stigliani & Ravasi, 2012). The outcome is a more meaningful problem when it aligns the entrepreneur's purpose with how society, technology, and culture are changing.

Our work contributes to entrepreneurship research in multiple ways. First, it adds to studies that view problems as the main entrepreneurial unit of analysis (Hsieh et al., 2007), and opportunities as valuable problem-solution pairings discovered through deliberate search or undeliberate recognition over a solution space (Nickerson et al., 2012; von Hippel & von Krogh, 2016). We speak to this literature by highlighting that problems not only allow alternative cognitive representations (Csaszar & Levinthal, 2016), but in the same way as solutions, can be designed and not just discovered. In providing conceptual and practical input for this design act, we respond to calls to bring problem framing and formulation center stage for creating novel value (Hyytinen, 2021; Nickerson et al., 2012).

The insights from our work are also fruitful for other disciplines in which problem solving is an established perspective to investigate the creation of novelty in organizational settings, for example, strategy (Baer et al., 2013) or innovation (Frishammar et al., 2016). Yet, studying reframing does not necessarily require adhering to a problem solving (and finding) tradition. This paper has drawn on concepts, including theories, visions, jobs, and paradigms (Argyris & Schon, 1974; Christensen et al., 2016), which as frames concern the overarching "why" of things, without using a problem solving vocabulary. We believe that future research on reframing should continue using this open-ended approach.

Our second contribution is to the literature that depicts entrepreneurs as theorists (Felin & Zenger, 2009; Wuebker et al., 2021), or more comprehensively as scientists (Zellweger & Zenger, 2021), who engage in theory building based on belief formation and associated theory testing. This latter phase has received a great deal of scholarly attention, particularly in the growing lean startup literature examining rapid, iterative prototyping and testing to validate the hypotheses (Camuffo et al., 2019; Blank, 2013). There is instead limited appreciation of the value of envisioning a theory beforehand, which fundamentally generates and motivates hypotheses amenable to testing

¹² A similar tool is the blue ocean canvas (Kim and Mauborgne, 2014), which visualizes differences between existing and new products in value parameters: the ones to be eliminated, reduced, augmented, or added to deliver a leap of meaning.

¹³ Discovery and design should be seen as conceptual polar types, here intended to illuminate their different natures. In reality, cases of problem finding exist that combine elements of both.

(Zellweger & Zenger, 2021). It is quite telling that Leatherbee and Katila (2020) reduce this early step to the number of hypotheses being formulated, while Eisenmann et al. (2013) consider it beyond the scope of their contribution to the lean startup process.

We believe that a narrow focus on experiments is not beneficial to entrepreneurship research and practice, which should occupy the influential center stage where why-related matters are tackled. Metaphorically speaking, our work on designing problems worth solving supports entrepreneurs in being more like Copernicus who advanced the heliocentric theory that started a scientific revolution (Kuhn, 1962) before being like Galileo who discovered evidence supporting Copernicus' theory. Specifically, we suggest reframing and the related concepts and tools (e.g., meanings, metaphors, scenarios) as important elements for theory building in the front-end of the entrepreneurial process. In fact, problem formulation and theory building are near-substitutes (Simon, 1996), as entrepreneurs actively compose theories by projecting their beliefs about a problem (Zellweger & Zenger, 2021). Our work advances and expands theory-based entrepreneurship research that remains rather generic as to what is theorized and how, and the anatomy of an entrepreneurial theory (Felin et al., 2021).¹⁴

Accordingly, it appears that sound principles and practices to build new theories might differ from, or even be orthogonal to, those for testing them. Imagination and assertion are critical to the former phase, whereas the latter relies on evidence-based analyses (Michewski, 2008). Consider the lean startup recommendation to try out ideas with users, no matter how early and rough they are. As new frames are based on the entrepreneur's contrarian beliefs, they are unlikely to be immediately understood by users. The risk is prematurely abandoning them in lieu of established frames that are more familiar to users, and thus less innovative. Should entrepreneurs delay feedback to when the problem underlying a belief has been thoughtfully framed? Should they fully form the vision in their mind's eye before submitting it for external evaluation? More research on these aspects is needed.

Third, this paper contributes to advancing research drawing on the core design tenets to investigate entrepreneurial phenomena (Berglund et al., 2020; Ding, 2019; Sarooghi et al., 2021). In portraying opportunities as emerging and evolving artefacts at the interface of organized individuals and their environment, this literature emphasizes solution-oriented artifacts including prototypes, canvases, pitches. We complement this work by drawing attention to more abstract, problem-oriented artifacts, such as frames, themes, jobs, thus providing detail to what Berglund et al. (2020) generally term opportunity visions.

In embracing the ideas of Schön and Dorst on the centrality of framing and reframing in design, we depart from most entrepreneurship studies that adhere to a Simonian view of design as search within a defined problem space (Berglund et al., 2020). This view is an important but incomplete input to problem finding. Dimov (2016) is an exception in arguing that framing the situation (i.e., "the imposition of a structure" on the problem) is an essential component of the entrepreneurial process, and that two entrepreneurs in the same situation would frame it differently. While Dimov (2016) recognizes problem frames as hidden, hard-to-observe working hypotheses that are refined and replaced, no detailed concepts and practical insights are offered to bring them to the surface and dynamically design them.¹⁵ Our paper is an early attempt to do so.

¹⁴ The recently proposed practitioner-oriented value lab approach (Felin et al., 2021) presents a canvas in which contrarian beliefs, a core problem, and a causal logic are key ingredients to formulate novel theories of value, yet remaining quite silent on how to come up with contrarian beliefs. Zellweger and Zenger (2021) suggest that, asking, e.g., "what important truth do very few people agree with you on?", and using the business model canvas to specify assumptions can help to some extent.

¹⁵ Our reframing approach is in line with the entrepreneurial design that Berglund et al. (2020) label transformation, a dialogical exchange of meanings across heterogeneous actors, relentlessly deframing and reframing problems

We believe that treating problems as design outcomes can advance the strategic impact of design and its role in entrepreneurship and management practice. Problems can be sources of differentiation: the inside-out connotation of problem design, driven by the entrepreneur's individual idiosyncrasies, can lead to more original frames. Ironically, the same biases that blur the vision of the problem according to user-centered design thinkers, contribute to designing the entrepreneur's own problem. Conversely, there is no special tie between a problem "out there" and the entrepreneur who discovers it. The problem is up for grabs and being the first to find it does not remove the risk of imitation. For instance, the fact that Lyft was the first ride-sharing company to address the problem of the underutilization of private citizens' cars by engaging non-professional drivers on its platform did not stop Uber from doing the same.

Still, problem design comes with complexities and risks. Its speculative nature requires the willingness to embrace discontinuity, ambiguity, and open-endedness, traits that define a design attitude to work (Michewski, 2008). When designing problems, there cannot be a moment of clarity, as in the case of entrepreneurs discovering a truth that was not part of their knowledge (Madsbjerg & Rasmussen, 2014). Indeed, when developing contrarian visions, the best an entrepreneur can achieve is a moment of less unclarity!

Will it be more difficult for entrepreneurs to be proficient at design thinking when it is not only about discovering and solving problems?¹⁶ Going forward, we need more research on the methods and tools that support effective reframing, especially in those areas that are distant from the traditional repertoire of user-centered design: hermeneutical phenomenology, deep reasoning, generative questioning techniques, metaphorical thinking, and world building.

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to maintain productive interpretive flexibility (Nambisan et al., 2017; Lainer-Vos, 2013).

¹⁶ As implied by the double diamond or IDEO approach to design thinking.

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